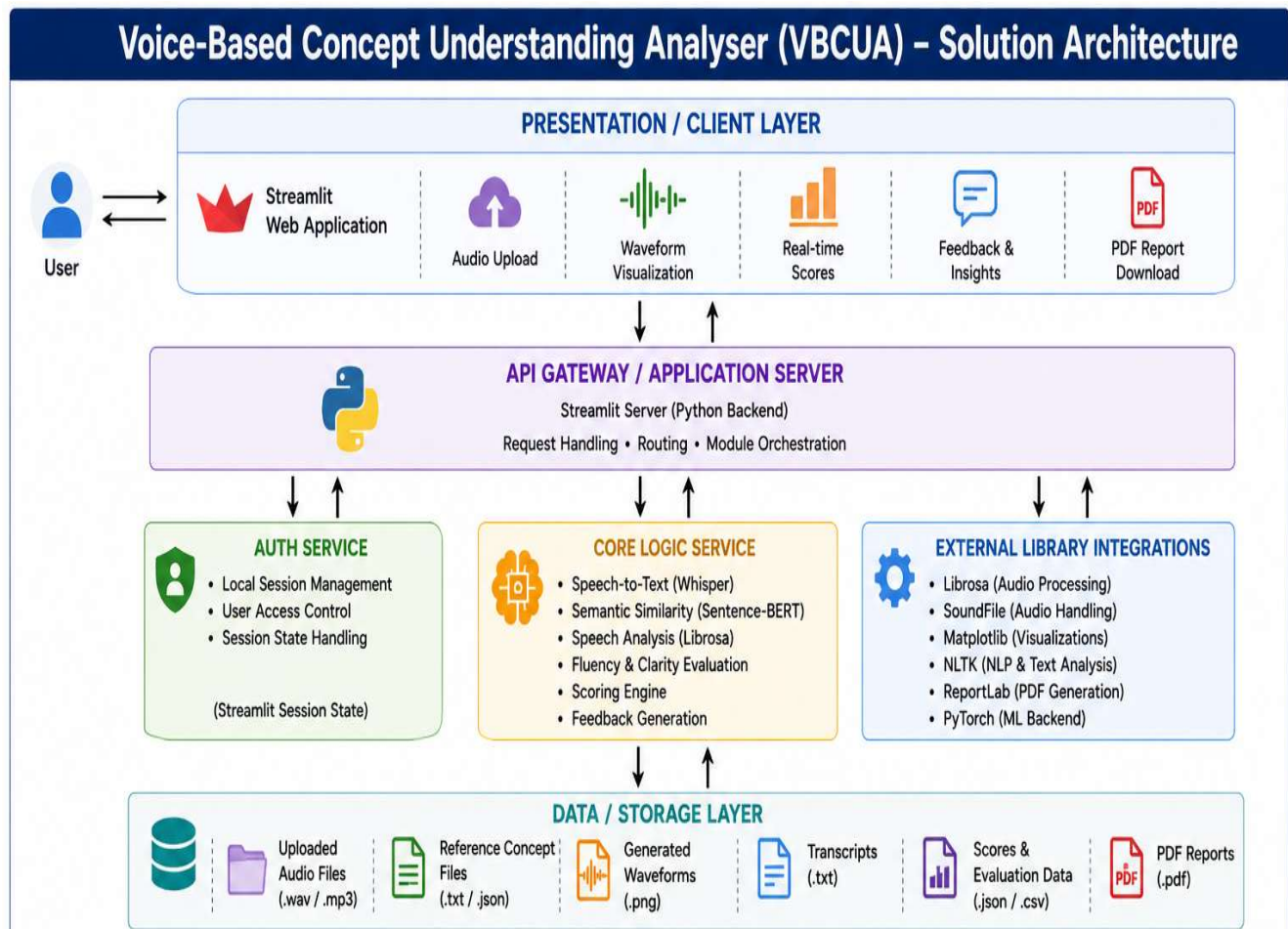


Solution Architecture

Date	1 July 2026
Team ID	
Project Name	Voice-Based Communication Understanding Analyzer
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides an interactive user interface for audio upload, playback, waveform visualization, concept evaluation, and report download.	Streamlit, HTML, CSS
API / Application Layer	Handles user requests, coordinates AI modules, processes uploaded audio, and manages application workflow.	Python, Streamlit
Authentication Service	Maintains local user sessions and application access (if implemented).	Streamlit Session State, Python
Core Logic Service	Performs speech transcription, semantic similarity analysis, speech feature extraction, comprehension scoring, and feedback generation.	OpenAI Whisper, Sentence-BERT, Librosa, PyTorch, NLTK
External Libraries	Supports visualization, report generation, and audio processing functionalities.	Matplotlib, ReportLab, SoundFile
Data / Storage Layer	Stores uploaded audio, generated transcripts, waveform images, evaluation scores, and PDF reports.	Local File System (Audio Folder, Reports Folder), TXT, PDF, PNG